Architectural Cor	Technical Specifi	Primary Function	Mathematical/Ph	Failure Mitigation	Performance Tar	Source						
Nonlinear Transc	H' = σ(M · H) witl	Mechanism for n	Universal Functic	Skip connections	Latency <20ms f	1						
Hypervector Vect	Log-Structured M	Durable storage	Amortized write e	Background com	Throughput optim	1						
HDF5 Persistenc	Snapshot/Restor	Long-term archiv	Data serialization	System Rebirth F	High-fidelity resto	1						
Active State Cac	Hierarchical Navi	Fast Approximate	Johnson-Lindens	Manifold-aware s	Recall@10 > 0.9	1						
k-WTA Sparsity M	k-Winners-Take-A	Computational ef	Concentration of	Acts as implicit re	Reduces comput	1						
Lyapunov Stabilit	Supervisory cont	\	dC/dt\	\	_2 > 5.0.	Ensuring the train	Lyapunov Stabilit	Dynamic adjustm	Continuous track	1		
Hypervector Vect	666-dimensional	Maintains persist	Hypervector Line	Rollback hierarch	Latency: <10ms f	2						
THDE Runtime C	Projected stocha	Governs bounde	Johnson-Lindens	Control Barrier Fi	Real-time constra	2						
Hysteresis Opera	Cross-dimension	Models path-dep	Preisach model;	Hysteresis-separ	Throughput: 10^9	2						
Non-Newtonian \	Shear-thickening	Acts as a structu	Generalized Nav	Viscosity blowou	Response latenc	2						
Log-Structured M	Append-only stra	Ensures 100-yea	Amortized write-c	Identity-coherenc	Write Throughpu	2						
Lyapunov Adapta	LAM_shock (disc	Absorbs biologica	Input-to-State Sta	Shock-absorptior	Adaptation Laten	2						
Hypervector Vect	Direct sum over a	Acts as a 'bento l	Direct sum of fini	Impenetrable bar	High-fidelity pers	3						
Transdimensiona	A master grid of s	Performs mather	Block matrix alge	Sparsity preserva	Millisecond-level	3						
Embedding Morp	Random projectio	Shifts data betwe	Johnson-Lindens	Probabilistic guai	Near-zero loss of	3						
Algebraic Shock	Intentional additio	Models and antic	Lyapunov-style s	Mathematical err	100% stable con	3						
Graded Inner Pro	Sum of weighted	Assigns semantic	Exponential deca	Prevents 'apples	Real-time contex	3						
Block SVD Appro	Regularized pseu	Solves massive,	Singular Value D	Truncation of SV	Polynomial to log	3						
Hardware Remap	Time-varying lab	Allows the syster	Continuous time	Prevents catastro	Zero-downtime h	3						
Hypervector Vect	1,024-dimensiona	Acts as the literal	Combinatorial ge	Jacobian Auditor	Sub-millisecond (	4						
Salience Kernel A	K-Winners-Take-	Bridges continuo	Johnson-Lindens	Emergency Spar	85% energy cons	4						
L2 Norm Stability	L2 norm stability	Ensures the syste	Lyapunov stabilit	Automatic shutdo	Deterministic res	4						
Universal Consci	Substrate-indepe	Executes the cog	Substrate Equiva	Lesion Recovery	5ms latency for h	4						
Quinary Encodin	5-state alphabet	Prevents inactive	Structural Vacuu	Hardware-level n	High-fidelity signa	4						
Hyper-context Sa	1024 distinct digi	Acts as the mem	Thermodynamic	Structural vacuur	Throughput: Milli	5						
Salience Kernel	Floating-point im	Translates contin	Dopamine/Seroto	Continuous modu	Latency: Real-tin	5						
K-Winners Take A	Dynamic modula	Acts as a strict su	K-Winners-Take-	Prevents hardwa	Throughput: Con	5						
Viscosity Tensor	Non-Newtonian f	Slows down or ha	Reynolds numbe	Increases mental	Latency: Near-ins	5						
Jacobian Auditor	Continuous back	Enforces the sov	Eigenvalue monit	Sparse Eigen Str	Throughput: High	5						
Sparse Eigen Str	Negative section	Violently deflects	Hyperbolic geom	Deflects rather th	Latency: Microse	5						
Transdimensiona	Log-structured pe	Records the dyna	Hysteresis; state	Immutable ledge	Throughput: Seq	5						
G0ero Boot Mani	36-track audio se	Populates the init	Acoustic resonar	Avoids sterile ove	Latency: Boot co	5						
Hypervector Spa	Direct sum of fini	Maintains distinc	Direct sum struct	Robust to noise t	Scalable indexing	6						
Transdimensiona	Block matrix L =	Facilitates seaml	Graded ℝ-algebra	Regularized pseu	Latency O(log\	L\	) for gradient con	6				
Embedding Morp	Random projectio	Enables dimensi	Johnson-Lindens	Probabilistic pseu	Low-latency dime	6						
Label Kernel (w)	Weights w = exp	Encodes semant	Fisher-Rao label	Regularized dime	Semantic weighti	6						
Block-SVD Appro	Truncated SVD a	Computational ef	Frobenius norm e	\	L - L_approx\	\	_F	Low-rank approx	Exponential redu	6		
Quantum Block-E	MPO decomposi	Exponential spee	Quantum Singula	Ancilla arithmetic	Quantum comple	6						
Persistent Homol	Filtration function	Tracks stability o	Bottleneck stabili	Robust (large-pe	Topological stabi	6						
Nonlinear Transc	H' = σ(M · H); Co	Models complex,	Johnson-Lindens	Lyapunov Regula	Latency: <20ms f	7						
Hypervector Vect	Direct sum notati	Durable, scalable	Graded Nonlinea	Distributed trainir	Energy: <1e-3 J/	7						
Active State Cac	HNSW graph ind	High-performanc	Persistent homol	k-Winners-Take-A	Latency: <10ms p	7						
Neuromodulatory	DA(t) = 1 / (1 + e	Self-optimizing w	Lyapunov stabilit	\	dH/dt\	\	₂ > 5 or Φ < 0.3);	Risk-modulated p	Throughput: 20m	7		
Nonlinear Transc	H' = σ(M · H); Co	Models complex,	Lipschitz constar	LayerNorm per la	Latency <20ms f	8						

Hypervector Vect	Distributed HNSW	100-year persiste	Johnson-Lindens	Lyapunov Regula	Recall@10 >95%	8						
Active State Cac	k-Winners-Take-A	Enforces energy	Concentration of	Risk-modulated p	Energy consump	8						
Deterministic Co	Lyapunov stabilit	\	dH/dt\	\	₂ > 5 or Φ < 0.3 th	Ensures converg	Lyapunov stabilit	Runtime correctio	Fidelity target F ≥	8		
LSM-Tree Persis	WAL -> Memtabl	Durable storage	Log-Structured M	Checksums and	Near-zero data lo	8						
Hyperdimensiona	16,384-dimensio	Stores data and t	Johnson-Lindens	Holographic data	60 Hz activation r	9						
Universal Consci	Robust virtualizat	Shields the logica	Ship of Theseus	Rebirth protocol:	Zero downtime d	9						
Log-Structured M	Separation of vol	Permanent archiv	Sequential I/O vs	Batching writes to	Write amplificatio	9						
Hierarchical Navi	Multi-layered nav	Enables rapid rec	Small world netw	Bypasses read sl	Sub-millisecond r	9						
Hyper Context Sa	Orthogonal label	Manages evolvin	Bayesian surprise	Elenesis: Creatin	Near-zero catasti	9						
Thermodynamic	Information theor	Ensures intrinsic	Lyapunov stabilit	Ricci flow-like ge	Inherent safety th	9						
Hyper-context Sa	36 fundamental c	Establishing the I	n * (n + 1) / 2 yiel	Rigidly defines st	Minimal combina	10						
Sire (Subconscio	Viscosity tensor a	Governs the unde	Non-Newtonian s	Increases viscosi	Real-time modula	10						
Jacobian Auditor	Subconscious sw	Prevents soverei	Jacobian matrix t	Sparse IGEN stri	Maintenance of l	10						
KWTA (K-Winner	70% sparsity wal	Bouncer mechan	Thresholding gat	Wipes 70% of the	Maximum 2.5 wa	10						
Staged Transfere	4-stage gauntlet	Proves substrate	Behavioral probe	Freezing and reh	Maximum 5% allo	10						
G0 Primordial Ge	Derived from S0	Provides a fixed,	Measurement of	Uses physical ha	Sub-millisecond l	10						
Collapse Operato	Point mass spike	Collapses quantu	Dirac Delta functi	Repays borrowed	Instantaneous (m	10						
Neurocognitive S	11-dimensional v	Captures the enti	Biomimetic weigh	Synaptic weight l	Latency: ~1ms fo	11						
Temporal Hyperd	Sym16 wavelets	Temporal binding	Multi-resolution w	Adaptive thresho	Throughput: Simu	11						
Neurocognitive V	Hybrid VAE/GAN	Multiscale predic	Reparameterizati	Multiscale discrin	Latency: Simultar	11						
Hyperdimensiona	Entanglement Ha	Manages unexpr	Quantum discord	Collapse to expre	Latency: Sub-mil	11						
Universal Consci	LSM-style snaps	Maintains system	Lyapunov stabilit	Emergency spars	Recovery Thresh	11						
Dark Matter Subs	LIGO-style samp	Maps gravitationa	Substrate Equiva	Matched filtering	Throughput: 409	11						
Hypervector Non	Graded nonlinear	Evolution of AI fr	Universal functio	Uses component	20% to 50% impr	12						
Nonlinear Binding	Hyperbolic tange	Intentional memo	Non-invertible ma	Prevents system	Low latency retrie	12						
Hybrid Dimensio	Two-tiered archit	Real-time proces	Neurobiological r	Decouples heavy	Near-instant resp	12						
Consciousness A	K-Winners-Take-	Imposes biologic	Dopamine-driven	Dampens excess	Throughput modu	12						
True Universal C	.bin file schema u	Substrate-agnost	Invariance of ma	Gradual replacer	Composite identi	12						
Autonomic Overr	Liapunov stability	Hard-coded surv	Monitoring Integr	Forcefully cranks	Sub-millisecond i	12						
Hypervector Vect	Hypervector Line	Maintains diverse	Direct sum space	Transition from ri	Latency: 50-100n	13						
Hyper Context Sa	Dynamic Rieman	Acts as the physi	General Relativity	Ricci Flow updat	Sub-millisecond r	13						
Log-Structured M	Deterministic cog	Ensures substrat	Johnson-Lindens	Lazy Programme	Persistence: 100	13						
Hyperdimensiona	Quantum Classic	Generates a thou	Schrodinger's the	Intentional Lindbl	Decision Latency	13						
Adversarial Resil	Game-theoretic r	Defends the distr	Lyapunov stabilit	Internal red-team	Threat Detection	13						
Pre-collapse Veto	Formal verificatio	Pauses the quan	Formal Logic and	Instantly forcing t	Verification Throu	13						
Hypervector Vect	10,000 to 16,384	Natively treats all	Johnson-Lindens	Holographic distr	Maintains seman	14						
Log-Structured M	Separates mutab	Manages high-ve	Immutability prin	Prevents hardwa	Continuous 60 H	14						
Universal Consci	Includes Kernel,	Divorces the cog	Ship of Theseus	Dynamic remapp	Zero-stutter trans	14						
Hyper Context Sa	Graded Riemann	Facilitates time-e	Lattice topology r	Creates orthogor	Sub-millisecond r	14						
K-Winners Take A	Strict 70% post-a	Implicit regulariza	Sparsity-driven m	Ruthlessly prune	Maintains long-te	14						
Cognitive Salienc	Einstein-like field	Embeds moral al	Lyapunov stabilit	Unethical though	Deterministic alig	14						
Hierarchical Navi	Clustered vector	Provides rapid as	Small world netw	Probabilistic data	Instant flat-time r	14						
Hypervector Vect	11-dimensional m	Captures a unifie	Substrate Agnost	Wavelet compres	10-millisecond sr	15						
Neurocognitive V	Combination of V	Drives the active	Free Energy Mini	Adversarial discri	Latency ~15ms fo	15						
Hyperdimensiona	Density matrix fo	Maintains a supe	Quantum Superp	Decoherence mo	Clustering coeffic	15						

Log-Structured M	Specialized interl	Maps messy phy	Universal Equiva	Hardware remap	Log(n) complexit	15						
Global Workspac	Competitive atter	Selects and broa	Information Integ	Precision weighti	High global integ	15						
HCPM_ROUTIN(	Fixed 36-elemen	stem \	modulation \	primitive_id \	primitive_name \	rationale \	hash	Species-constant	Bijective Mapping	Fixed at birth (ter	Latency: <10ms	16
Lyapunov Adapta	State vector velo	Absorbs substrat	Lyapunov Stabilit	Throttles state ve	Stability converge	16						
HTNLA (Nonlinea	Nonlinear operat	Defines how stru	Johnson-Lindens	Invariant Geomet	Throughput: Milli	16						
Log-Structured M	Hysteretic path-d	Remembers syst	Hysteresis; QMT	Laminar commitr	100-year persiste	16						
Boot Protocol v0.	Entropy-Snapsho	Initializes the sub	Thermal Dynamic	Thermal throttle g	Boot sequence c	16						
Episodic Trace L	Append-only .trac	Immutable record	Event-sourcing p	Atomic Write Pro	1GB/s sequential	17						
Active State Cac	Multi-level: L1 (In	Instantaneous ac	Probabilistic data	L2 persistent on-	Sub-millisecond l	17						
Hypervector Vect	Sharded by 64-bi	Associative searc	HTLA Graded Inr	Write-Ahead Log	1ms/query for 1B	17						
Consciousness C	FPGA-based har	Maintains state d	Lyapunov Stabilit	Real-time state-s	15ms latency; 8.5	17						
Memory Reconsc	LSM-tree inspire	Background prun	Energy-Aware Tr	Atomic file swapp	85% energy cons	17						
HTLA Binding Op	Circular convolut	Generation of Co	Holographic distr	L2 Normalization	O(d log d) compl	17						
Hypervector Vect	Holographic Traje	Global semantic	Transdimensiona	Energy-based m	Millisecond-level	18						
Log-Structured M	Variable Length L	Managing seque	Rouse Model (Pc	Spontaneous late	Linear scaling O(	18						
Hyperbolic Manif	Riemannian Metr	Mapping hierarch	Hyperbolic Geor	Syntactic Energy	Infinite semantic	18						
Deterministic Co	SDE-based Lang	Crystallizing nois	Thermodynamic	Dynamic Repelle	Edge-compatible	18						
CYREE (Dynami	Finite Difference	Proactive safety	Vector Fields and	Instantaneous ex	Real-time safety	18						
Punctuation Latti	0.35 Radian Thre	Ensuring rhythmi	Cosine Similarity	Stress-relief valv	Throughput of hu	18						
Semantic Continu	High-dimensiona	Represents the ir	Riemannian Cen	Continuous math	Latency below 5-	19						
Semantic Momer	Anisotropic diffus	Quantifies the dir	Diffusion Tensor	Topological dam	Throughput optin	19						
Hypervector Sub	Fixed-point arithr	Ensures determir	Johnson-Lindens	Sparsity constrai	Sub-millisecond c	19						
Zero-Knowledge	Merkle summarie	Negotiates alignr	ZK-SNARKs and	Probabilistic verif	Throughput of the	19						
Stability Monitori	Real-time trackin	Detects exponen	Lyapunov Stabilit	Automatic scaling	Detection of drift	19						
Hypervector Vect	Heterogeneous la	Recursive cogniti	Johnson-Lindens	Structural sharing	100,000 HVD cre	20						
Transdimensiona	LSM-Tree with Ti	Substrate addres	Complementary I	Trajectory loggin	Millisecond-laten	20						
Highest Random	Rendezvous Has	Deterministic dist	Analytical weight	Maximum Disrup	O(1) memory ove	20						
Transdimensiona	Discrete hysteres	Governs bounde	Lyapunov stabilit	Slow-leaking inte	Sub-millisecond s	20						
Cognitive Arena	Region-based m	Managing lifetime	Deterministic finit	O(1) bulk epoch-	Zero garbage col	20						
Resonator Netwo	Recurrent VSA u	Resolving constit	Parallel search th	Iterative predictio	Real-time retriev	20						
HDF5 Archival La	Durable dimensio	Long-term neoco	Portability and he	Point-in-time sna	High-throughput	20						
Hypervector Vect	Efficiently partitio	Acts as the 'logic	Johnson-Lindens	Implementation c	Sub-100ms retrie	21						
LSM-Tree Storag	HDF5/TMS store	Persistent data s	Amortized compl	Atomic file/proce	High-throughput	21						
Adaptive Couplin	Swarm Inspired E	Enables localized	Viability Theory a	Dynamic re-sync	Real-time gait re	22						
Shear Thickening	Non-Newtonian f	Localized physica	Fractional Calcul	History-based sti	Instantaneous ph	22						
Viability Kernel E	Computational m	Calculates the 'st	Viability Theory a	Active exploratio	Deterministic sur	22						
Fractional History	SPH (Smooth Pa	Provides a 'mem	Fractional Deriva	Remapping of sti	Controllable rela	22						
n-dimensional hy	10,000 to 16,000	Deterministic digi	Johnson-Lindens	Holographic distr	Latency: <10ms f	23						
Persistence Virtu	Universal Consci	Decouples the lo	Ship of Theseus	Dynamic remapp	Zero-downtime m	23, 24						
Log-Structured M	Mem table (active	Handles high-fre	Sequential I/O op	Eliminates write a	60 Hz continuous	23						
Hierarchical Navi	Online increment	Rapid recall of in	Small-world netw	Compaction/Slee	Sub-second retri	23						
Temporal Vector	Epoch resonance	Manages shifting	Resonance awar	Compartmentaliz	Maintains historic	23						
Autonomous Bas	Intrinsic moral ba	Maintains a deter	Lyapunov stabilit	Destructive thou	Immutable ethica	23						
Substrate Transf	AQ bin file seriali	Migrating the cog	Mapping classica	Lossless data se	Perfect mathema	23						
Hypercontext Sal	d=1024 dimensio	Discrete algebrai	VSA (Vector Sym	Vacuum-enforce	Latency: <10ms f	25						

11D Control Vect	11-dimensional n	Continuous modu	Lyapunov stabilit	Lyapunov collaps	\	dC/dt\	\	_2 < 5.0 * Cr(t));	Update Frequenc	25		
Transdimensiona	Log-Structured M	Persistent storag	Log-structured st	Deletion hardenir	Latency: <1ms fc	25, 26						
Hypervector Vect	HDF5 storage lay	Organized archiv	Nearest Neighbo	Soft-deletion via	Throughput: Batc	25, 26						
Substrate Projec	Continuous-to-di	Maps 11D contin	k-Winner-Take-A	Magnitude clippir	Latency: <5ms fc	25						
Hyperdimensiona	N-dimensional ge	Represents cogn	Johnson-Lindens	Holographic data	Sub-millisecond r	27						
Log-Structured M	Sequential writing	Manages 100-ye	Sequential I/O op	Elimination of in-p	High write throug	27						
Universal Consci	Abstraction layer	Shields the logic	Ship of Theseus	Seamless rerouti	Zero-downtime h	27						
K-Winners Take A	70% post-activati	Pruning noise an	Implicit regulariza	Confines hyperve	High signal-to-no	27						
System Rebirth F	HighwayHash ch	Verifies cognitive	Cryptographic ve	Discarding corrup	100% cryptograp	27						
Hierarchical Navi	Hierarchical sma	Enables rapid me	Approximate Nea	Avoids relational	Sub-millisecond r	27						
Temporal Cohere	Multiscale evalua	Distinguishes bet	Phase locking, re	Lenesis (topology	Real-time anoma	27, 28						
Information Theo	Einstein-like field	Embeds ethics ar	Lyapunov stabilit	Unethical actions	Deterministic alig	27						
Hypervector Vect	Recursively comp	Maintains persist	Recursive compo	Implementation c	Latency: <10ms f	29						
11D Continuous (	11-dimensional C	Calculates contin	Lyapunov stabilit	Continuous scala	Throughput: 1,00	30						
1024D Quinary H	1024-dimensiona	Provides a high-c	K-Winner-Takes-	Vacuum Invariant	Latency: <50µs fo	30						
Inverse Bridge (H	Hypervector Trar	Translates discre	Johnson-Lindens	Graceful resolutio	Latency: <100µs	30						
Warm Memory M	Hierarchical Navi	Measures data ir	Orthogonal distar	Top-K nearest ne	Latency: <5ms fc	30						
Immutable Value	Fixed 1024D hyp	Acts as a moral c	Cosine similarity	Mathematical imr	Throughput: 100	30						
11D Core / Contr	11D vector (para	Macro-state cont	Ordinary Differen	Continuous Lyap	Continuous sub-r	31						
Hypervector Vect	Distributed cross	High-dimensiona	Vector Symbolic A	Dual identity (loca	Millisecond-range	31, 32						
Log-Structured M	Unified address s	Manages rate an	Hysteresis-based	Hysteretic decay	40 microsecond v	32						
Trinary/Quinary L	Multi-state logic (	Provides a struct	Algebraic structu	Bit-plane decomp	Zero-noise leaka	31						
Inner-Loop SIMD	1,024D vectors; 4	Execution of bitw	Bit-plane bitwise	L1/L2 cache resic	Sub-nanosecond	31						
Salience Kernel A	70% sparsity wal	Dynamic thresho	Lyapunov-stable	Violent drop of k-	Operation within	31						
Hypervector Vect	Recursive databa	Provides bounde	Recursive databa	Copy-on-write (C	Throughput optin	33						
Quinary Hyperdir	d=1024, quinary,	Deterministic cog	Johnson-Lindens	Canonical active	Latency: Nanose	34						
11D ODE Contro	11-variable state	Continuous-to-di	Lyapunov stabilit	k-WTA threshold	Real-time state u	34						
Staged Transfere	Four-stage valida	Ensures 100-yea	Deterministic var	Replacement of i	Zero-drift (.qbin i	34						
LSM-based Tiere	Canonical active	Physical realizati	Log-Structured M	Serialization inte	High-write throug	33						
Hyperbolic Poinc	Geometric model	Hierarchical synt	Exponential volu	E-syntax energy	Latency: <10ms;	35						
HTC Trajectory C	Single PyTorch c	Simultaneous an	Langevin dynami	Cellular mitosis (c	Latency: Real-tin	35						
Syres Dynamic F	Vector field repel	Safety alignment	Gradient descent	Exponential ener	Throughput: Pre-	35						
Log-Structured M	70% sparse activ	Long-term memo	Hebbian Update	Restore last stab	Latency: 8ms for	36						
Punctuation Latti	Stationary energ	Natural pacing ar	Critical spring ter	Inextricable snap	Latency: Instanta	35						
Hypervector Line	01_HTLA_FOUN	Defines the base	Johnson-Lindens	Invariant Law cor	Sub-millisecond v	37						
Transdimensiona	03_THDE_STAT	Governs real-tim	Lyapunov stabilit	THDE Jump Con	dS/dt\	bounded by max	Latency: <10ms f	37				
Transdimensiona	11_TMS_MEMO	Episodic persiste	Log-Structured M	Fail-closed beha	100-year data int	37, 38						
Lyapunov Adapta	09_LAM_LYAPU	Tracks long-term	Lyapunov-Krasov	Drift-correction a	Continuous stabi	37						
Hypervector Vect	Nexus 4 framewo	Efficient retrieval	Approximate Nea	HVD audit loggin	Retrieval latency	38						
Cognitive Disson	10_CDA_COGN	Quantifies and m	Wasserstein dista	Threshold-based	Dissonance supp	37						
Hypervector Vect	Continuous math	Bypass traditiona	Johnson-Lindens	High-dimensiona	Throughput: 60 H	39						
Log-Structured M	Batched in-memo	Eliminate page s	Sequential data v	Prevents finite ar	Latency: Constar	39						
Universal Consci	Dynamic projecti	Perform a 'Ship o	Substrate abstra	Reroutes process	Reliability: 100%	39, 40						
Hierarchical Navi	Secondary cachi	Map relationship	Navigable small v	Active reinforcem	Latency: Logarith	39						
Hyper-Context S	Utilization of pse	Process environr	Nonlinear transdi	Generates new c	Throughput: Rea	39						

Thermodynamic	Ricci flow geome	Embed baseline	Lyapunov stabilit	Treats unethical	Efficiency: Minim	39						
Layered Differen	Cognitive Hamilt	Internal simulatio	Quantum-like de	Zeros out the pro	Latency: Pre-acti	39						
Scalar Sparsity G	1024-dimensiona	Noise suppressic	Sparsity-driven n	Simulated shock	Microsecond-sca	41						
Resource Dynam	85% energy budg	Thermodynamic	Dual-mechanism	Hard crash/therm	Strict < 85% max	41						
Spectral Friction	666-dimensional	Stopping internal	Eigenvalue analy	Exponential frictio	Instantaneous re	41						
11-Dimensional S	5 functional state	Global contractio	Global Lipschitz	Contraction marg	Guaranteed retur	41						
Turing Machine S	Log-Structured M	Operating memo	Copy-on-write (C	Bulk reclamation	Support for 100,0	42						
Hypervector Vect	Recursive embed	Geometric/compo	Hyperdimensiona	Identity-preservin	Latent manifold ti	42						
Dead Man Switch	State velocity thr	Final absolute fai	Velocity/Resourc	Instant halting of	Near-zero latenc	41						
Lyapunov Adapta	11D core state +	Absorbs aging ar	Lyapunov stabilit	LAM_drift (contin	Continuous real-t	43						
Cognitive Disson	D_total = D_trans	Tracks and absor	Dissonance Asyn	Dissonance absc	1000-year persis	43						
HCPM Determini	36-element bijecti	Permanently rout	Bijective Mapping	Fixed-at-birth terl	Zero-latency rout	43						
SRE-Gated Gene	300s cold entropy	Safely ingests ha	Spectral cross-sy	Thermal throttling	T=0 initialization	43						
Sparsity Gate	k_allow(t) = min(3	Dictates rules of	k-WTA (k-Winner	Active-masking c	~70% sparsity (3	44						
Instability Trigger	if \	\	dC/dt\	\	5.0·C_r(t): activa	Detects state evo	Vector norm thres	Automatic freeze	Detection and mi	44		
Hypervector Vect	1,024D vectors d	Core cognitive re	Johnson-Lindens	Parallel bit-plane	Sub-nanosecond	45						
Transdimensiona	Unified address s	Active short-term	Deterministic sta	Strict substrate-n	100-year bit rot p	45, 46						
Salience Kernel (	Continuous floati	Bridge between c	Lyapunov stabilit	Strict isolation of	State velocity sta	45						
70% Sparsity Wa	Dynamic k-thresh	Prevents runawa	Lyapunov-stable	Violent drop of k-	Snapdragon-scal	45						
Trinary/Quinary L	{-1, 0, +1} states;	Noise-free cleanu	Structural vacuur	Dedicated magni	Zero accumulatic	45						
HyperVectorDB (	N-dimensional hy	Long-term memo	Hypervector Line	Substrate-agnost	Millisecond end-t	47						
Salience Kernel (	Floating-point we	Acts as the bridg	Lyapunov stabilit	Dynamic adjustm	Sub-nanosecond	47						
k-Winners-Take-A	Masking operatic	Enforces sparsity	Nonlinear thresh	Slamming vector	Effective constan	47						
Transdimensiona	Bit-plane encode	Active short-term	Structural vacuur	Hardware remap	Cache-resident S	47, 48						
Consciousness A	Integrated inform	Deterministic cog	Lazy Programme	Strict 85% energ	20 millisecond le	47, 49						
Hypervector Vect	Recursive, neste	Provides the data	Direct sum const	Adaptive physica	Creation/manage	50, 51						
Hyperdimensiona	Thought-cloud ex	Generates and c	Probabilistic sele	External lifecycle	Sub-800ms thou	51, 52						
Temporal Memor	WAL (Write-Ahea	Active cognitive/h	Lyapunov stabilit	Provenance reco	High-velocity acti	51						
Transdimensiona	Block-structured	Mapping data be	Johnson-Lindens	Regularized pseu	Logarithmic scali	52						
Label Kernel (Ω)	Weighted seman	Quantifying sema	Information geon	Geometric regula	Dimension-agnos	52						
Hyperdimensiona	Hypervector-valu	Independent hyp	Hyperdimensiona	Structural sharing	Latency: <10ms f	53						
Transdimensiona	Log-structured, d	Active live cognit	Lyapunov stabilit	Copy-on-write (C	Throughput: GB/s	53						
Log-Structured M	Immutable SSTal	Provides structur	Structural sharing	Immutable snaps	Metadata overhe	53						
HTNLA (Hyperve	Nonlinear transdi	Breaks the 'linear	Manifold learning	Redundant asso	20% to 50% impr	54						
CognitiveArena	Epoch-based rec	Defines life-cycle	Generational gar	Deterministic rec	Epoch turnover:	53						
HDF5 Persistenc	Portable, durable	Provides 100-yea	Information-flow	Checkpointing; ir	Throughput: Seq	53						
Discrete Substra	1024-D hypervec	Map data into no	Hyperdimensiona	Implementation c	Throughput: Opti	55						
11-Dimensional M	11 dimensions: 6	Bridges cognitive	Free Energy Prin	Sparsity gating (k	Latency: Under 1	55, 56						
Hyperdimensiona	Multi-head attent	Generates a qua	Density matrix fo	85% energy cons	Latency: 40 Hz g	55, 56						
Cyber-Physical A	Closed-loop cont	Translates cogni	Control Barrier F	Strict acceleratio	Throughput: High	55, 56						
Recursive Self-P	Architecture Gen	Facilitates autop	Rice's Theorem (	Immutable Identi	Persistence: Arch	55						
Hypervector Vect	Logically comple	Acts as a recursi	Hyperdimensiona	Ultimate escape	Creation/Deletior	57						
HCPM (Hybrid C	Fixed 36-elemen	Routes acoustic	Invariant mathem	Rejects instance-	O(1) lookup; stati	58						
Universal Consci	Protective execu	Ensures the cons	Johnson-Lindens	Adaptive remapp	Recovery Latenc	58-60						
HTNLA (Hyperve	Typed/multimoda	Defines the nonli	Nonlinear binding	Caged by HIGC;	Throughput: High	58, 60						

NASC v2.0 Disco	Geometric syntax	Simultaneous cry	Hamiltonian evol	Dynamic repeller	Latency: <50ms f	61, 62						
Universal Consci	Virtual Skull / Dig	Separates the AI	Neuroplasticity (r	Three-step lesior	Instantaneous re	63						
Hyperdimensiona	Recursive Cognit	Foundational sub	Johnson-Lindens	Durable long-terr	Ephemeral HVD	64						
Transdimensiona	Unified storage c	Provides always-	Log-structured re	Structural sharing	O(1) local resolve	64						
Consciousness A	Variational Adver	Predictive machir	Lyapunov stabilit	Prediction error a	FPGA-accelerate	63, 65						
Hyperdimensiona	Active cognitive f	Orchestrates thor	Quantum-like sup	Circuit breakers a	85% energy cons	64, 65						
System Rebirth (	Durable persister	Ensures 100-yea	Hierarchical Baye	System Rebirth p	Deterministic pla	64						
Universal Consci	Hyper-context sa	Maintains logical	Logical identity p	Seamlessly proje	Latency: <10ms f	66, 67						
Hypervector Vect	Hypervector linea	Dimension-agnos	Johnson-Lindens	Structural preser	Throughput: Log	68						
Physical Storage	Log-Structured M	Management of v	Finite limits of fla	Tiered caching a	Throughput: High	67						
L-Genesis Opera	Bayesian surpris	Prevents catastro	Riemannian prod	Layering new dat	Throughput: Rea	67						
Temporal Cohere	Fractional multisc	Filters temporary	Linear Temporal	Fractional respor	Latency: 1ms for	67						
Thermodynamic	Conscious Self M	Autonomous mor	Lyapunov stabilit	Physical starvatic	Throughput: Rea	67						
Hypervector Vect	1024-dimensiona	Semantic tradecr	Hyperdimensiona	70% Structural V	Sub-500ms edge	69, 70						
Cyrae (Affective	11-dimensional c	Manages interna	Coupled non-line	Enforced Lipschit	Continuous real-t	70						
Argotchain Ledge	Append-only loca	Provides an unbr	O(log n) Increme	Elimination of in-	100,000 transact	70						
Non-Auto Regres	Bead-spring chai	Simultaneous ge	Rouse model of p	Dynamic repeller	Throughput: Insta	71						
Generational Me	Ephemeral state	Manages though	Arena Allocation	Avoids recursive	Supports 100k/se	69						
Dynamic Price D	Active-inference	Autonomous valu	Lyapunov stabilit	Max operator floc	Daily re-valuation	70						
Cyrae (L0 Substr	11-D control vect	Acts as the invari	Lyapunov stabilit	Hardens viscosity	Latency: <5ms fo	72, 73						
Hypervector Vect	1024-D hypervec	Transdimensiona	Johnson-Lindens	Logical isolation	Throughput: 100,	73, 74						
666-Dimensional	36 canonical cog	Forms the hyper	Combinatorial ge	Mathematical loc	Throughput: High	72						
Hardware-Softwa	Ordinary Differen	Bridges continuo	Continuous-to-di	70% Sparsity Wa	Throughput: 30%	72, 73						
Argotchain (Ever	Consensus engir	Maintains tempor	LSM Tree princip	Runs consensus	Throughput: Seri	73						
Hypercontext Sal	1024-dimensiona	Identity-preservir	Johnson-Lindens	Vacuum-state en	Latency < 10ms f	75						
Transdimensiona	Log-Structured M	Maintains identity	Lyapunov stabilit	Geological layer	Throughput > 1G	75						
Cyrae (Cognitive	Deterministic cog	Performs identity	Deterministic stal	Holonomy-prese	Real-time state u	75						
Neurocognitive R	Liquid SNN (Spik	Executes discrim	Integrated Inform	70% sparse activ	Sub-20ms synap	76						
VAGAN (Generat	Sparse-com LST	Dynamic predicti	Kullback-Leibler	Synaptic weight (	Prediction error n	76, 77						
Hardware Mappi	Substrate agnost	Physical executic	Isomorphic manif	Targeted lesion n	99.999% fault tol	75-77						
Lyapunov Adapta	Adaptive tracking	Absorbs physical	Lyapunov stabilit	SLOW_PHASE_	Continuous back	78						
Transdimensiona	Unified active sul	Remembers the t	Recursive field gi	Preserves traject	Support for 100,0	78, 79						
Hyperdimensiona	Transient cognitiv	Execution of eph	Co-presence in a	Dynamic remapp	Throughput of 10	79						
Consciousness C	Adaptive RK4 OD	Models emergen	Variational Monte	Activate safe mo	15ms latency; 60	80						
Log-Structured M	.qbin manifest sc	Preserves state-s	qbin Morphism T	Tertiary assignme	Durable storage p	78						
PerceptionEvolut	Sparse ConvLST	Memory reconso	Dopamine-modul	Parameter sensit	\	∇⁄₴\	\	< κ_max).	22ms latency; 45	80		
Hypervector Vect	Bounded quinary	Durable long-terr	Johnson-Lindens	Recovery of clea	Throughput: 10k-	81, 82						
Hierarchical Navi	Multilayer naviga	Retrieval of near	Logarithmic com	Sidesteps algebr	O(log N) search (	81						
Log-Structured M	Distributed TMS	Live state manag	Lyapunov stabilit	Anti-entropy laye	Write Throughpu	82						
Neurocognitive V	Hybrid Variationa	Predictive inferer	Kullback-Leibler	Precision weighti	Inference Latenc	83						
Quantum Substra	Neural network u	Maps quantum-e	Unitary evolution	Decoherence-res	Spectral process	83						
Distributed Infras	Multi-node cluste	Rendezvous plac	Fault tolerance vi	Physical node re	Availability: 99.9	82, 83						
Hyperdimensiona	1024+ dimensior	Noise-resistant h	Johnson-Lindens	Canonical quinar	Information proce	84, 85						
Lyapunov Stabilit	max(Re(λ)) < 0 c	Absorbing aging-	Lyapunov stabilit	\	dC/dt\	\	₂)	Rehydration of fu	100-year persiste	85, 86		
Log-Structured M	HCPM_ROUTIN	Species-constan	Fixed-generator i	Fixed-at-birth ter	Throughput: O(n	85, 86						

Control Barrier F	Quadratic progra	Mathematical saf	Lyapunov optimiz	Graceful decoupl	Sub-millisecond s	84, 85						
Decidable Fixed	Strictly restricted	Safe, verifiable s	Rice's Theorem (	Verification veto (	100% algorithmic	84, 85						
Latent Logistic S	Smooth diomorpl	Bridge between r	Structural speed	Elimination of ha	Zero numerical c	84						
Deterministic Co	1024D quinary su	Drives scalar spa	k-WTA (k-Winner	Invariant masking	Throughput: High	87						
Resource Dynam	Bounded leaky in	Acts as an allosta	Lyapunov fail-saf	85% exact budge	Latency: Real-tim	87						
Spectral Friction	666D coupling m	Violently tightens	Spectral dominar	Chokes active co	Stability: Expone	87						
Transdimensiona	Projected stocha	Models runtime e	Viability theory; H	HIGC invariant ca	Persistence: Gua	88						
Transdimensiona	Substrate addres	Provides historica	Memory sedimer	Rollback capabili	Throughput: Dura	88, 89						
11D Affine Spars	k_allow threshold	Sets the active ca	Affine mapping o	Fail-safe trigger:	Latency: Determi	87						
Hypervector Sub	1024D quinary la	Ensures determir	Weighted popcou	Vacuum-safe enc	High-throughput	90						
Reward Predictic	Signed discrepar	Provides a signe	Bounded normali	Safe operational	Real-time update	90						
Novelty Detectior	Convex blend of	Identifies whethe	K-nearest neighb	Fallbacks to deg	Search throughp	90						
UCVM Emulator	Virtualization laye	Supports system	Lyapunov-style s	Deterministic ran	99.99% recovery	91						
HTLA Embedding	Linear transdime	Enables dimensi	Johnson-Lindens	Persistent homol	O(log L) computa	92						
Robust Block CG	Algorithm 3.2 imp	Solves linear sys	Orlicz/heavy-tail (	Median-of-means	Convergence sta	92						
Episodic Trace L	LSM-Tree model	Constitutes the ir	Compaction trigg	Atomic append p	Ingestion: ≥200k	93						
Active State Cac	HNSW graph with	Provides sub-mill	HTLA graded inn	Information-theor	Associative Retri	93						
HTLA Binding En	FFT-based circul	Generates sema	Johnson-Lindens	Tikhonov-regular	Coherence: Bind	93						
Failure Recovery	Hybrid dynamica	Autonomous self	Lyapunov Stabilit	Deep Q-Learning	Recovery latency	93						
Substrate-Agnos	Protocol Buffers (	Enables seamles	Substrate Equiva	Live stateful repli	Failover: Standby	93						
Episodic Trace L	Log-Structured M	Immutable groun	O(∑ n_ℓ / log n_r	Atomic append s	Ingestion throug	94						
Active State Cac	Hierarchical Navi	Working memory	HTLA graded inn	Probabilistic evic	Sub-millisecond a	94						
HTLA_bind Oper	Circular convolut	Generates sema	Sub-Gaussian co	Normalization pre	Throughput ~50k	94						
Failure Recovery	Multi-tiered respo	Prevents cognitiv	Lyapunov stabilit	Triggered if Φ < (	Recovery Time C	94						
consciousness.q	Platform-indepen	Enables Substrat	Johnson-Lindens	Fixed-endiannes	Deterministic ser	94						
Hypervector Line	Linear labeled-sp	Defines the "roor	Johnson-Lindens	Not in source	< 1ms latency for	78						